# Kulachart Parnduangkeaw



*Machine Learning & Data Science · Full-Stack Web & Mobile · Computer Engineering, SWU*

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**Top 30** KBTG KAMPUS Hackathon 2026

**Top 24** NTT DATA Digital Innovation Challenge 2026

**0.89 AUC-ROC** skin-cancer classifier, 2026

**~300 users** SWU Run web app, 2026

## Summary

Computer Engineering student at SWU (expected 2028) and 2026 Data Science Intern at Botnoi Group who builds ML systems end to end, from model to app. Seeking an AI / ML, Data Science or Full-Stack internship.

## Experience

**Data Science Intern** — Botnoi Group · May – Jul 2026

- Built product demand-forecasting models for SME clients, including feature engineering that improved model accuracy.
- Set up the MLOps pipeline to train and deploy the models, and built dashboards for the client companies.

## Projects

**KaTang Cash-flow Co-pilot** — KBTG KAMPUS Hackathon 2026 · Team of 3 · Data science · AI

- Built the gradient-boosting spending forecast and a Monte Carlo simulation of 4,000 thirty-day cash-flow paths (simulated data) behind a safe daily spending limit for variable-income first jobbers; pitched in the top-30 round.

  Python · scikit-learn · Pandas · Machine Learning

**WELLYNC** — NTT DATA Digital Innovation Challenge 2026 · Team of 3 · Web & LLM dev · AI · Web

- Built an Amazon Bedrock LLM agent with role-scoped tool calling that drafts chronic-care plans, saved only after a doctor confirms, on serverless AWS (Lambda, DynamoDB, Cognito); reached the top 24 teams.
- Built the Next.js web app where patients, caregivers and doctors share one consent-checked record.

  Next.js · TypeScript · Tailwind CSS · AWS

**Derma AI: Skin Cancer Screening** — CPE310 Healthcare AI · Solo project · 2026 · AI · Web

- Reached 0.89 AUC-ROC and 88.7% top-2 accuracy on 3,619 unseen test images with the best of the three models; exported to ONNX so it runs in the browser and images never leave the device.
- Fine-tuned ImageNet-pretrained ResNet50, EfficientNet-B3 and DenseNet121 on ISIC 2019 (25,331 images, 8 classes) with class-weighted loss for the imbalanced classes.
- Split by lesion (StratifiedGroupKFold) so no test lesion is seen in training; tuned each model's cancer threshold on validation to target 90% sensitivity.

  Python · PyTorch · scikit-learn · Pandas

**Peak Electricity Day Alert** — CPE312 Data Science capstone · 2026 · Data

- Forecast a week ahead which days hit peak demand from 2015–2020 Panama load and weather data, with leakage-safe daily features and a time-based train/test split.
- XGBoost reached **84% recall on peak days vs 73%** for a same-day-last-week baseline, at higher precision (78% vs 73%); also compared Logistic Regression.

  Python · XGBoost · scikit-learn · Pandas

*More projects*

**Smart Flood AI** — DOH Hackathon 2026 · Team of 3 · Web & AI dev · AI · Web. Zero-shot Grounding DINO pipeline that finds cars and wheels in highway CCTV to estimate water depth, plus the Next.js dashboard.

**SWU Run Together 2026** — SWU Athletics Club · Team of 2 · Developer · 2026 · Web. React web app on live Google Sheets that gives runners a digital BIB with their order status; **used by about 300 runners**.

**BrainFit** — Client project · Team of 5 · 2025 · Mobile. Built the sign-in flow (Apps Script + Google Sheets) and some screens of a React Native (Expo) brain-training app commissioned by nursing students; **delivered to the client for their research**.

**Employment Data Gap Analysis** — Research project · 2025 · Data. Random Forest on 100,000 placement records vs a survey of 80 CPE students: students over-rated internships and portfolio (**31% vs 24%** model weight).

**Prim's Algorithm Visualizer** — CPE101 course project · 2024 · Web. D3.js tool that builds a Minimum Spanning Tree step by step.

**This Portfolio** — Web App · 2026 · Web. One typed data file drives kulac.dev and this CV, checked in CI on every push.

## Leadership & Teaching

**Computer Engineering Student Leader** — Dept. of Computer Engineering, SWU · 2024 – Present

- Represent Computer Engineering students at faculty meetings and university events as the main link between students and faculty.
- Plan and run department workshops and student-development programs.
- Taught micro:bit programming to high-school students at a faculty activity and emceed a university event (2025).

**Robotics Teaching Assistant** — Srinakharinwirot University · 2026

- Taught robotics and programming fundamentals to high-school students in hands-on sessions.
- Prepared and maintained the robotics equipment and ran live robot-programming demos.

**Metaverse Presentation Assistant** — Srinakharinwirot University · 2025

- Assisted with a lecture on Metaverse technology (how VR, AR and AI work together in virtual worlds) and built its interactive slides and live demos.

**Research Presentation Assistant** — Dept. of Computer Engineering, SWU · 2025

- Helped a faculty member present a research poster at Chulachomklao Royal Military Academy on classifying Acanthamoeba cysts in microscope images with pretrained CNN features and an SVM classifier.

## Education

**Bachelor of Engineering in Computer Engineering** — Srinakharinwirot University (SWU) · 2024 – Present · Expected 2028

Relevant coursework: Software Development, Database Systems, Artificial Intelligence, Healthcare AI Systems, Computer Architecture, Electronics

## Honors & Certifications

- Top 30, KBTG KAMPUS Hackathon 2026 (KaTang Cash-flow Co-pilot)
- Top 24, NTT DATA Digital Innovation Challenge 2026 (WELLYNC)
- Bangkok Insurance Scholarship — Bangkok Insurance, 2025
- Google Cloud, Big Data & GenAI Training — Google Cloud, 2026

## Skills

**AI / ML:** Machine Learning, Deep Learning, Computer Vision, Generative AI, PyTorch, scikit-learn, XGBoost, OpenCV, Grounding DINO, ONNX Runtime, Amazon Bedrock (LLM agents, tool calling)

**Methods:** Demand forecasting, time-series forecasting, transfer learning, time-based validation, threshold tuning, Monte Carlo simulation, MLOps

**Programming:** Python, SQL (MySQL), Java, C/C++

**Data:** NumPy, Pandas, Data Analysis, Matplotlib, Seaborn, Google Sheets, Google Colab

**Web:** React, Next.js, TypeScript, JavaScript, Tailwind CSS, D3.js, HTML/CSS, Vite, Framer Motion

**Mobile:** React Native, Expo

**Backend & cloud:** FastAPI, AWS (Lambda, API Gateway, DynamoDB, Cognito), Firebase, Google Apps Script, Google Cloud

**Tools:** Git, GitHub Actions, Vercel, Hugging Face Spaces

**Languages:** Thai (native), English (intermediate)

Scan for live demos & project write-ups

**kulac.dev**